

Exam 1 Review

Question 1

A large logistics company in the US wanted to examine the delivery performance of its drivers. Data were collected from 100 randomly sampled drivers with 45 of them being younger drivers (Group A) and 55 older drivers (Group B). The drivers operated across southern, central, and northern regions of the country. On average, younger drivers completed deliveries in 4 days, while older drivers (Group B) averaged 6 days. In addition to delivery times, the company also recorded the driver's gender (male/female/other), how long each driver had been employed (in years) and the average weight (in tonnes) of the packages they delivered.

- a. Classify this scenario as an *observational* or *experimental* study. Give a reason your classification.
- b. Write two possible *research questions* that the company can answer using the data collected in this scenario.
- c. Identify the *sample* and the *population* of interest.
- d. What are the *cases/observational units* in this study? How many are they?

- e. Identify a suitable parameter of interest and the associated statistic in this study.
- f. List ***all*** the variables in the study and identify each variable as numerical or categorical. For each numerical variable state whether it is discrete or continuous. For each categorical variable state whether it is nominal or ordinal.
- g. Identify a variable (or variable set if more than one) that can be reasonably summarized using:
- a simple bar plot
 - side-by-side box plots
 - a histogram
 - a scatter plot
 - a stacked or side-by-side bar plots
 - a contingency table

Question 2

Below are weights (in pounds) of 5 randomly sampled patients who visited St. Joseph's Clinic on a certain day. Assume that this sample, though small, is representative of the patients that visit this clinic.

182, 186, 177, 185, 183, 181

- Calculate the mean, median, and quartiles (upper and lower) for the weights. Show/explain your work. Do you think the mean would be a reliable measure of the typical weight of the patients that visit this clinic? Why or why not?
- Calculate the *standard deviation* for the weights. Show/explain your work. What does this standard deviation tell you about the weights?
- Classify the statistics you calculated in parts (a) and (b) as robust or non-robust. Explain your reasoning.

d. Suppose a nurse notices an error in the data entry. The weight entered as 186 lbs should have been 168 lbs. Without doing any calculations, state whether each of the following statistics will increase, decrease, or stay the same if the 186 is changed to 168. Explain your reasoning.

- mean
- median
- standard deviation
- interquartile range
- Range